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PHYSICS

The Symmetry That Hid Itself

Nothing in the equations of the weak force permits a particle to have mass, and the whole edifice was rescued by allowing empty space to have a preference.

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The bottom of a wine bottle is pushed up into a dome, and a marble dropped onto the dome will not stay there. It rolls off in some direction and settles into the circular trough, and the direction it picks is meaningless — every point on the trough is as good as any other. But once the marble has chosen, the situation is no longer symmetric. An observer arriving late, finding the marble at rest, would see a system with a preferred direction and no obvious reason for it.

Pierre Curie put the principle the other way round in 1894: the symmetries of a cause should appear in its effects. It is an appealing rule and it is false, or at least it fails in the one way that matters most. A system's equations can be perfectly symmetric while none of its stable states are. Physicists call this *spontaneous symmetry breaking*, which is a slightly grand name for the observation that the marble has to land somewhere.

Iron is the everyday example. The forces between the atomic magnetic moments in a lump of iron do not care which way is north; the interaction depends only on the angle between neighbouring spins. Above 1043 kelvin the lump obliges and shows no net magnetisation. Below it, all the moments line up in some direction, and the direction is an accident of history. The law is rotationally symmetric and the magnet is not.

This turns out to be the reason anything has mass.

The Massless Nuisance

In 1960 Yoichiro Nambu, thinking about superconductivity in the language of quantum field theory, noticed that the broken-symmetry state carries a signature [1]. If a continuous symmetry is broken, there must exist excitations that cost arbitrarily little energy — because moving the whole system a little way along the trough costs nothing at all, and a long-wavelength wave that does so locally costs almost nothing. In a magnet these are spin waves, and the flatness of the trough shows up directly in their dispersion: the energy of a long-wavelength magnon falls off as the square of its wavevector, which is why a ferromagnet loses magnetisation as $T^{3/2}$ at low temperature instead of exponentially. In a relativistic field theory the same modes are particles with exactly zero mass. Nambu and Giovanni Jona-

Lasinio built a model of the strong interaction around the idea the following year [2], and Jeffrey Goldstone turned the observation into a theorem [3], proved in general with Abdus Salam and Steven Weinberg in 1962 [4].

For particle physics this looked like a disaster rather than an opportunity. The weak interaction was known to be short-ranged, which meant its carriers had to be heavy; but a gauge theory of the kind Sheldon Glashow had written down in 1961 [5], with its symmetry group of $SU(2) \times U(1)$, insists that gauge bosons be massless. Adding a mass term by hand destroys the gauge invariance and with it any hope of a sensible theory. Breaking the symmetry spontaneously seemed the natural escape, and Goldstone's theorem slammed it shut: instead of heavy force carriers you got a fresh crop of massless particles that nobody had ever seen.

> **WHY SHOULD BREAKING A SYMMETRY PRODUCE A MASSLESS PARTICLE AT ALL?**

The reason is worth dwelling on, because it is the same reason the escape route exists. The trough at the bottom of the wine bottle is flat. Excitations that move along it have no restoring force, and no restoring force means no mass. The theorem is really a statement about the geometry of the ground state, not about any particular field.

The Superconductor Loophole

The way out had already been found, in a laboratory rather than on a blackboard. In 1933 Walther Meissner and Robert Ochsenfeld discovered that a superconductor does not merely fail to resist current; it actively expels magnetic field from its interior [6]. Two years later Fritz and Heinz London wrote down the electrodynamics of this, and the field turned out to die away exponentially inside the material over a characteristic distance [7]. For niobium that distance is about 39 nanometres [8].

An exponentially screened field is precisely what a massive force carrier produces. Reading the London penetration depth λ_L as a Compton wavelength gives

$$m_\gamma = \frac{\hbar}{\lambda_L c} \approx \frac{197.3\ \text{eV nm}}{39\ \text{nm}} \approx 5.1\ \text{eV},$$

so inside a block of niobium the photon weighs about twice the energy of a green light quantum. The superconductor has spontaneously broken a gauge symmetry, and rather than producing a massless Goldstone boson, it has produced a massive photon.

Philip Anderson pointed this out in 1963, in a paper of four pages [9]. His argument was that when the broken symmetry is a *gauge* symmetry, the would-be Goldstone boson and the massless gauge field are not two separate problems but one solution. A massless spin-one particle has two polarisation states; a massive one has three. The Goldstone mode supplies the missing longitudinal state. It does not vanish and it does not appear as a separate particle: it is absorbed, in the standard and unimprovable phrase, *eaten*.

Four Months in 1964

Anderson's argument was non-relativistic, and whether it survived a proper relativistic treatment was genuinely unclear. Three groups settled it within a single autumn.

François Englert and Robert Brout, at the Université Libre de Bruxelles, submitted first, in June; their paper appeared in *Physical Review Letters* on 31 August [10]. Peter Higgs, in Edinburgh, sent a short note to *Physics Letters* in July, which was accepted [11], and a second and fuller one about a week later, which was not. The rejection is the reason the Higgs boson has a name. Revising the rejected paper for *Physical Review Letters*, Higgs added a sentence pointing out that an essential feature of this kind of theory is the prediction of incomplete multiplets of scalar and vector bosons — that is, a leftover massive scalar particle [12]. He later remarked that the reference would never have appeared in print had the first journal taken the paper. The referee who handled it for the new journal, as Higgs learned only two decades afterwards, was Nambu.

Gerald Guralnik, Carl Hagen and Thomas Kibble published the third treatment in November, with the cleanest account of how the massless pole cancels [13]. Higgs wrote the argument out at length in 1966 [14], and Kibble generalised it to non-Abelian groups in 1967, showing how the mechanism can leave one gauge boson massless while making the others heavy [15] — which is exactly the arrangement nature uses, since the photon survives while the W and Z do not.

Weinberg put the pieces together into a model of the weak and electromagnetic interactions of leptons in 1967 [16]. It attracted essentially no attention for four years, on the reasonable grounds that nobody knew whether such a theory could be calculated with at all. Gerard 't Hooft proved in 1971 that it could [17], and the citations arrived. Englert and Higgs shared the 2013 Nobel Prize; Brout had died two years earlier.

The Arithmetic of an Empty Space

What the mechanism requires is a field with a potential shaped like the bottom of the bottle. Write ϕ for the field and

$$V(\phi) = -\tfrac{1}{2}\mu^2\phi^2 + \tfrac{1}{4}\lambda\phi^4,$$

with $\mu^2 > 0$, so that $\phi = 0$ is a maximum and the minimum sits at $\phi = v = \mu/\sqrt{\lambda}$. That value v is the vacuum expectation value: the amount of field that empty space contains when nothing else is going on.

The satisfying part is that the theory is not free to choose it. The muon's lifetime fixes the Fermi constant to half a part per million [18], and the Fermi constant fixes v outright:

$$v = \left(\sqrt{2}\,G_F\right)^{-1/2} = 246.22\ \text{GeV}.$$

The mass of the Higgs boson is the curvature of the potential at the bottom of the well, so $m_H^2 = 2\lambda v^2$, and the measured 125.20 GeV [19] gives $\lambda = 0.1293$. Two measurements, both parameters, no freedom left. The figure below plots that potential to scale.

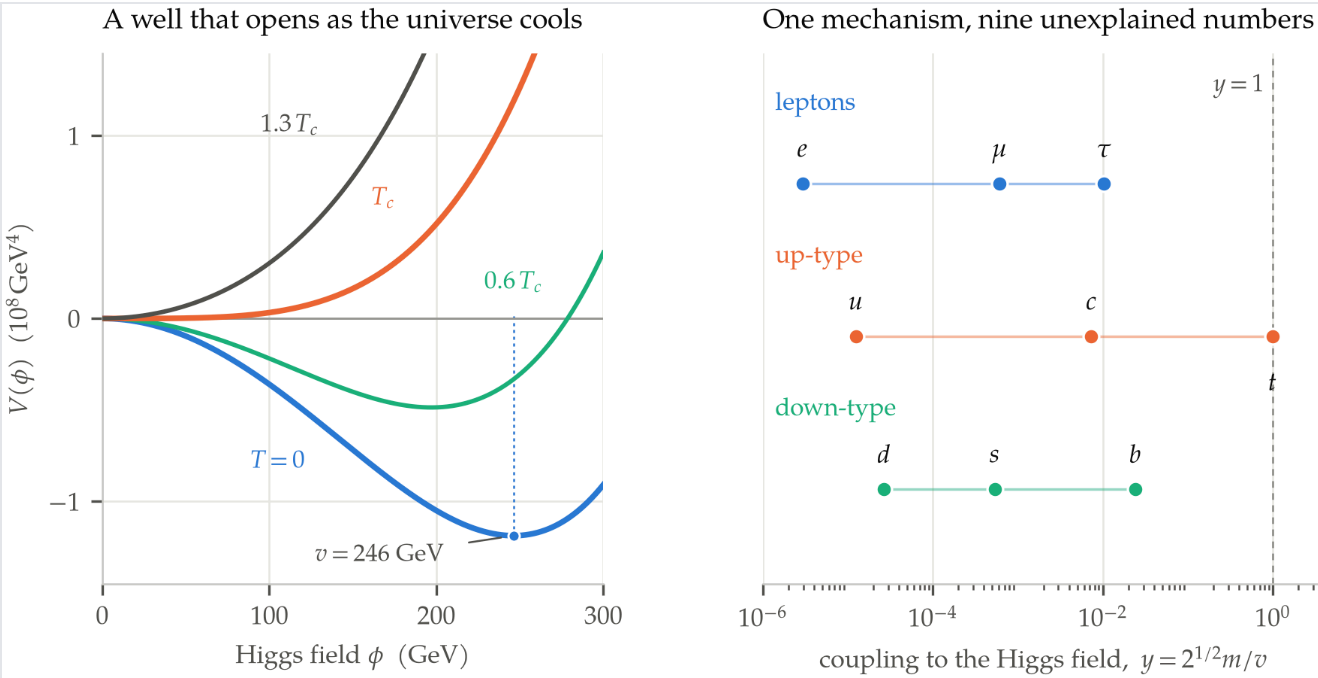

FIGURE The Higgs potential fixed by two measurements, and the couplings it does not fix. Left: the tree-level potential with v and λ set by the Fermi constant and the Higgs mass, warmed by a leading thermal mass term normalised so the quadratic term vanishes at the lattice crossover temperature. In the Standard Model the change of shape is a smooth crossover rather than the sharp transition a bare Landau potential suggests. Right: every charged fermion's Yukawa coupling, spanning five and a half decades.

The gauge bosons then get their masses from how strongly they couple to that resident field. For the W this is $m_W = gv/2$, which inverts to a gauge coupling $g = 0.6528$. A better test is that the same breaking pattern predicts a relation between the two boson masses and the weak mixing angle. Taking the two measured masses,

$$\sin^2\theta_W = 1 - \frac{m_W^2}{m_Z^2} = 1 - \left(\frac{80.3692}{91.1880}\right)^2 = 0.2232,$$

against a directly determined on-shell value of 0.22342 [19]. Three independent measurements agreeing to one part in a thousand is the kind of thing that makes a mechanism believable.

The residual is interesting too. Continuing at tree level, the electric charge should be $e = g\sin\theta_W$, which implies a fine-structure constant of 1/132.1 where the measured low-energy value is 1/137.036 — a discrepancy of 3.6 per cent. Alexander Sirlin organised these corrections into a single quantity in 1980 [20]. Two effects dominate it: the running of the fine-structure constant up to the Z mass, worth about 6 per cent on its own, and a top-quark loop pulling back by roughly 3 per cent. The top's contribution grows as m_t^2, and that quadratic sensitivity is why electroweak fits could weigh the top quark years before the Tevatron made one in 1995.

Ingredient	Before breaking	After breaking
Scalar doublet	4 real fields	1 Higgs boson
Three SU(2) bosons	6 polarisations	9 (W and Z)
One U(1) boson	2 polarisations	2 (photon)
Total	12	12

The bookkeeping in the table is the clearest statement of what “eaten” means. Nothing is created or destroyed; three of the four scalar fields are reassigned as the longitudinal polarisations of three gauge bosons that did not previously have any. One scalar is left over, and in July 2012 the ATLAS and CMS collaborations found it [21], [22]. A decade of further data then confirmed the part that makes the mechanism falsifiable rather than merely elegant: the strength with which the new particle couples to everything else scales with that thing’s mass, in every channel measured precisely enough to check [23].

What the Mechanism Does Not Do

It is worth being precise about the claim that the Higgs field gives things mass, because the popular version of it is wrong by about two orders of magnitude.

The Higgs field does set the masses of the elementary particles, through couplings $y = \sqrt{2}\,m/v$ that the theory does not predict. The right-hand panel of the figure shows the damage: the electron’s coupling is 2.9×10^{-6}, the top quark’s is 0.991, and the intervening seven charged fermions are scattered across five and a half decades of a logarithmic axis with no discernible pattern. The mechanism explains how a mass can exist in a gauge theory. It says nothing whatever about why the top quark is 337,000 times heavier than the electron, and the fact that the top’s coupling is one to within a per cent has never been explained either.

More strikingly, almost none of the mass you are made of comes from this. A proton contains two up quarks and a down quark whose current masses total about 9 MeV, which is under 1 per cent of the proton’s 938 MeV. A lattice decomposition of the proton’s mass into the terms of the QCD energy-momentum tensor puts the quark condensate contribution — the only part that responds to the Higgs field at all — at 9 per cent, with the remaining 91 per cent split between the kinetic energy of quarks, the energy in the gluon field, and the trace anomaly [24]. Ordinary matter weighs what it weighs because of the strong interaction. The Higgs field is a rounding error on it.

> **IF THE HIGGS FIELD WERE SWITCHED OFF TOMORROW, WOULD THE W BOSON BECOME MASSLESS?**

It would not, and the reason is the best joke in the Standard Model. The strong interaction breaks its own chiral symmetry spontaneously, and the pions are the resulting Nambu-Goldstone bosons. Those pions carry exactly the right quantum numbers to be eaten by the W and Z. Leonard Susskind pointed out in

1979 that in a world with no Higgs field, QCD alone would still break the electroweak symmetry, giving the W a mass of $gf_\pi/2 \approx 30$ MeV [25]. Chris Quigg and Robert Shrock worked through such worlds in detail in 2009 [26]. The mechanism is right; the scale is wrong by a factor of $v/f_\pi = 2673$. What the Higgs field supplies is not the idea of electroweak symmetry breaking but the size of it.

The Coldest Thing That Ever Happened

Symmetry breaking is a low-temperature phenomenon. Heat the iron past 1043 kelvin and the magnetisation goes; heat the universe enough and the Higgs field's preference goes the same way, because thermal fluctuations add a positive term to the effective mass and flatten the well. Lattice simulations put the changeover at $T_c = 159.5 \pm 1.5$ GeV [27], which is 1.9×10^{15} kelvin, reached about 9 picoseconds after the beginning.

Whether that changeover was a phase transition depends on the Higgs mass, and here the answer arrived before the measurement did. In 1996 Kajantie, Laine, Rummukainen and Shaposhnikov showed non-perturbatively that the line of first-order transitions ends somewhere below 95 GeV [28]. At 125 GeV there is no transition at all — no bubbles, no latent heat, no boiling early universe, just a smooth crossover during which the vacuum quietly acquires its preference. That is an inconvenient result, because a first-order transition would have supplied the out-of-equilibrium conditions needed to generate the excess of matter over antimatter. The Standard Model cannot do it, and the crossover is one of two independent reasons why not; the other is that the CP violation available in the quark mixing matrix falls orders of magnitude short of what the observed matter excess requires.

There is a final oddity in the numbers. Run the Higgs self-coupling λ up to very high energies and it decreases, driven mainly by the enormous top-quark coupling, and at somewhere around 10^{11} GeV it crosses zero — meaning the potential turns over and our vacuum is not the true minimum but a very long-lived metastable one [29]. The measured Higgs and top masses land close to the boundary between stability and instability, close enough that the sign depends on inputs known to a per cent. Nobody knows whether that near-criticality is a clue or a coincidence.

What is not in doubt is the shape of the argument. A theory that forbade mass entirely was rescued by allowing the vacuum to be less symmetric than its own laws, an idea imported wholesale from the physics of magnets and superconductors and then confirmed, forty-eight years later, by finding the one leftover particle the bookkeeping demanded.

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